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Strategic Transformation of Gas Breakthrough Control Optimization: Toward a Scalable Automated Optimization Framework

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Abstract

This paper discusses the challenges of establishing an effective process and management system for oil reservoir optimization. It focuses on two of Petroleum Development Oman's (PDO) largest oil fields, where Gas Breakthrough Control (GBC) technology has been applied to sustain low-GOR oil production from Gas-Oil-Gravity-Drainage (GOGD) wells. Initially, GBC delivered substantial benefits, including reduced gas production, sustained net oil output, decreased flaring. However, performance deteriorated over time due to lack of structured management, increased well count and hardware constraints. This paper presents PDO's ongoing efforts to sustain and enhance GBC performance year-round, highlighting the recent transformation of GBC optimization work from a reactive campaign- based approach to a proactive, data-driven, and ultimately digitized system, capable of managing long-standing challenges. The transformation directly supports PDO's mission of closing the gap to potential through unscheduled deferment reduction, flare control, and oil optimization. Additionally, the paper outlines future strategies to scale and enhance the effectiveness of GBC across broader operational contexts.

Introduction

As production systems become increasingly complex, optimization has remained a critical strategic focus in the upstream oil and gas sector. In response to this challenge, Petroleum Development Oman (PDO) introduced an innovative technology to improve the operation of Gas-Oil-Gravity-Drainage (GOGD) production wells in two of its largest oil fields approach in late 2007.

The initiative was centered around the adoption of Gas Breakthrough Control (GBC) technology, which enables the sustained operation of GOGD wells at their target potential while maintaining a low Gas-Oil Ratio (GOR) - which is not achievable through conventional fixed choke operations^[1]. The complex production behavior of GOGD wells, where the same choke position produces different production results, has often been attributed to complex or unpredictable reservoir drainage dynamics, and it has led to frequent gas breakthrough, leading to flaring and substantial oil deferment due to facilities gas handling constraints. The GBC control provide an automated feedback control which adjusts the well head choke control valve to maintain a constant differential pressure across a fixed orifice.

The principle behind this approach, is that the differential pressure attribute to any rudimentary bulk volumetric flow control, where the parameter to be controlled is affected by both produced liquid and produced gas flow rates^[1] as depicted in Fig.1. Through statistical well testing, optimal differential pressure setpoints can be established, enabling appropriate optimal production targets.

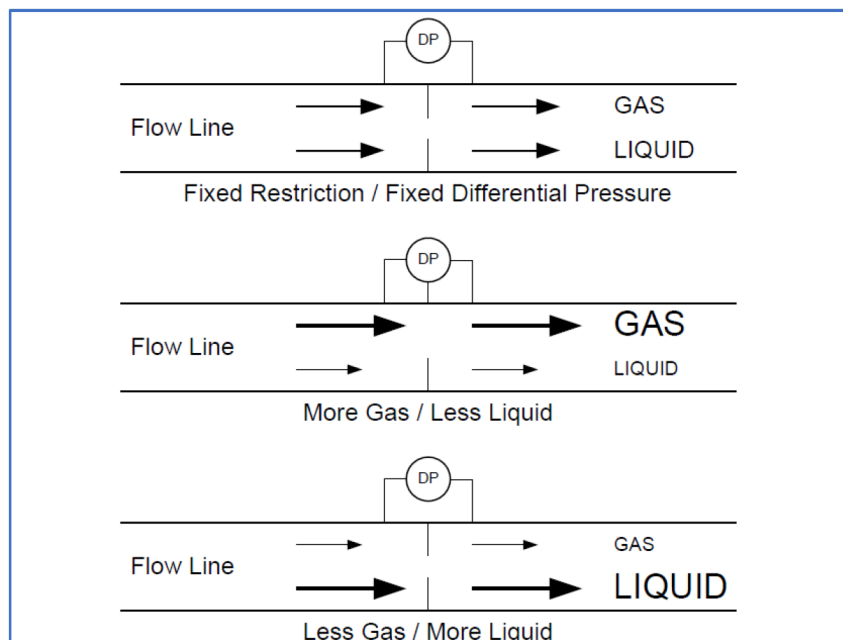


Figure 1—Closed-Loop Control adjusts the well head choke control valve to maintain a constant differential pressure across a fixed orifice.

As soon the GBC technology has been rollout early 2007, it has proven to be effective, delivering substantial optimization benefits and contributing to overall production efficiency. Now days, with more than 400 GBC-equipped wells in operation, the technology has successfully sustained production contribution levels of approximately 13% over the past two decades.

However, the challenges associated with GBC implementation has been demonstrated with sustainability of the performance, primarily stemmed from the increased number of GBC wells, and the need for continuous evaluation of control settings and regular review of gas lift targets.

To address these challenges and ensure sustained GBC performance, a dedicated short term GBC optimization campaigns were launched since 2013 through collaboration between the Function and the cluster team. The campaigns have yielded positive results in the short term, with focus on hardware maintenance and technology improvements such as using different type of control valves. Following campaigns have yield key improvements included the development of comprehensive Standard Operating Procedures (SOPs) and improve in the monitoring. Fig.2 shows the standard optimization process for each well.

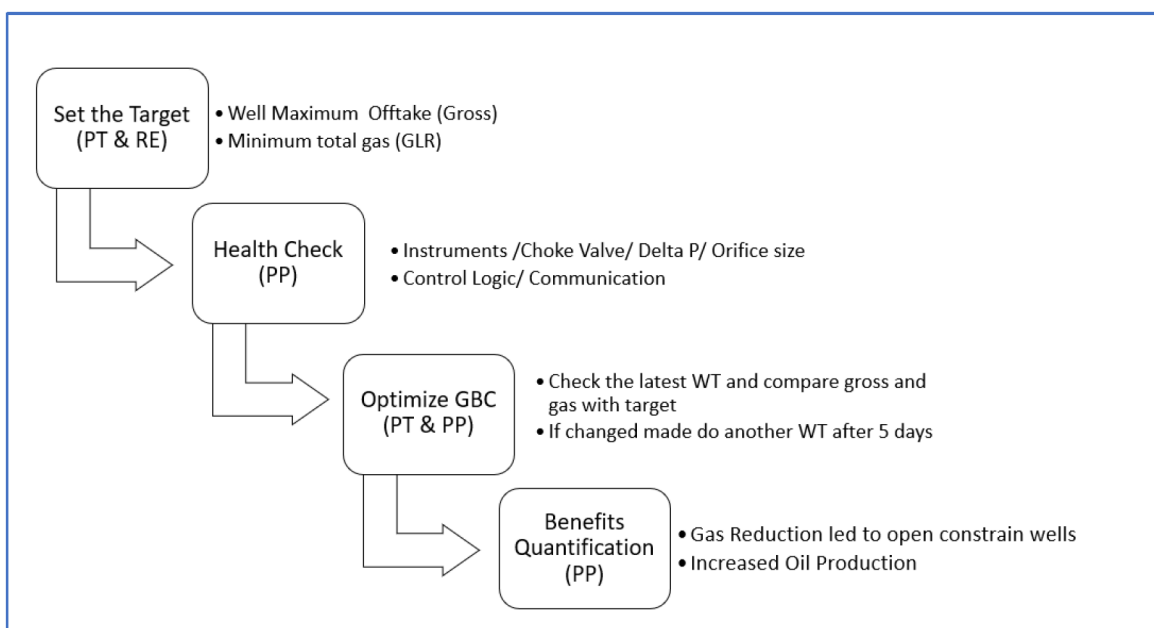


Figure 2—2018 GBC Optimization SOP

However, the optimization campaigns continued to rely heavily on trial-and-error approaches and individual expertise, leading to inconsistencies in well optimization. As a result, some wells were not fully optimized, resulting in elevated gas production resulting in rising the GOR, increased facility constraints, and subsequent oil deferments. Fig. 3 illustrates this upward GOR trend, largely influenced by reservoir behavior, where the performance of GBC plays a pivotal role in maintaining reservoir parameters.

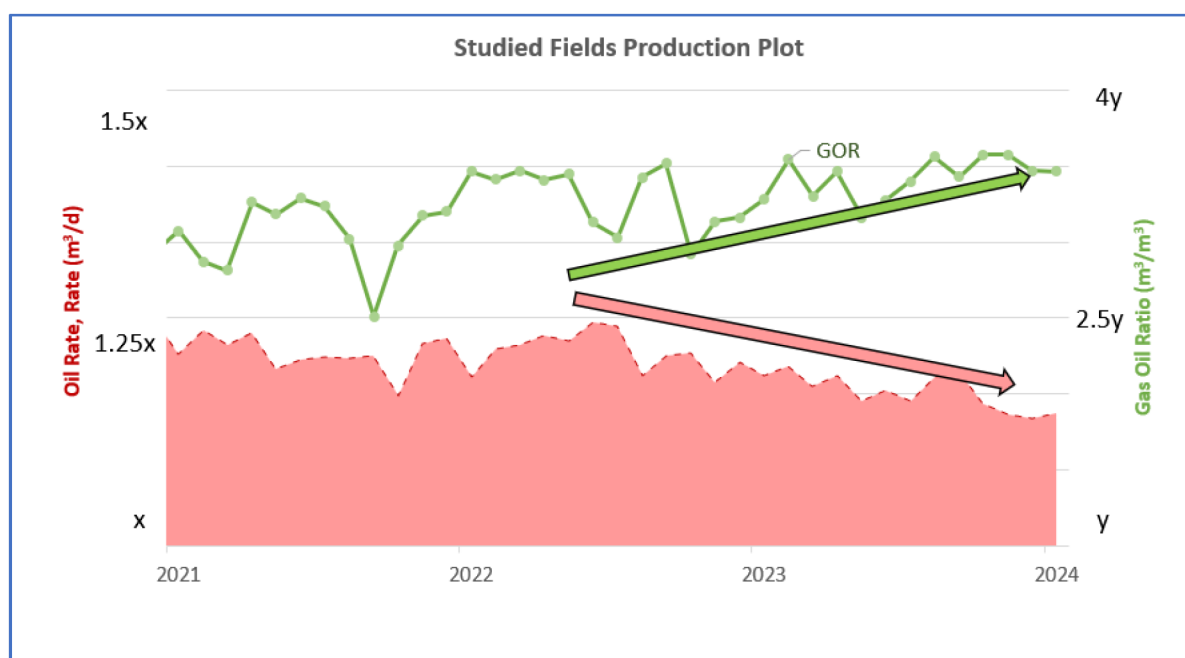


Figure 3—Drop in oil production with increase in GOR in the field.

The following section explores the key operational sustainability challenges, their root causes, and the analytical methods used to address them.

Operational sustainable Challenges

Although re-optimization campaigns have been successful in the short term, long-term performance improvements have been elusive, necessitating repeated intervention.

In the 2024 optimization campaign, a comprehensive review of GBC technology and performance was conducted. This analysis identified two primary categories of operational sustainability challenges, as outlined below.

Increasing Number of GBC wells

Over the past 15 years, the number of GBC wells in the studied fields has tripled as shown in Fig.4. Projections indicate that the number of GBC wells will increase by 20% by 2030. This rapid growth poses a significant challenge to the traditional, manual optimization model, which requires dedicated people effort for each individual well. As the system scales, the current approach becomes increasingly unsustainable.

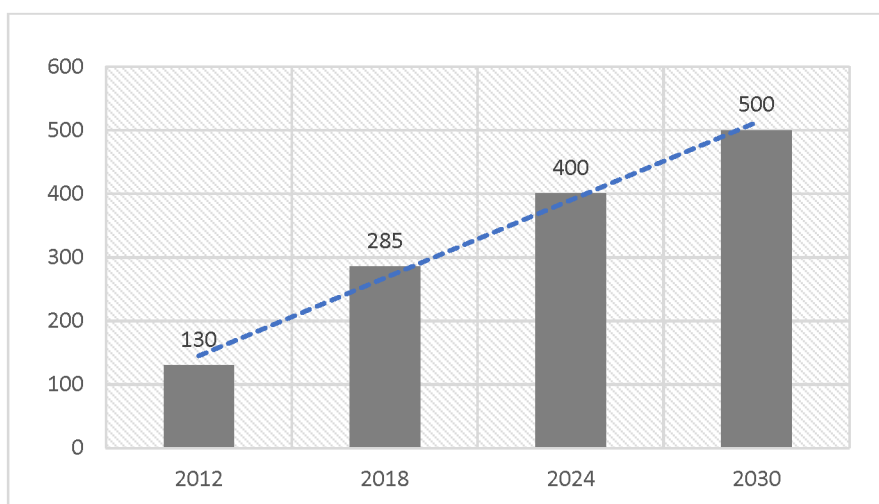


Figure 4—Number of GBC wells over the years.

Lack of Management System

Following the initial rollout and post-implementation reviews, GBC operations continued without a clearly defined management structure or ownership across stakeholders. This has demonstrated in several recurring issues:

Delay in detecting GBC hardware resulting in value loss. The hardware components of the GBC technology are a main pillar of the GBC performance. A failure of any components can lead to bad outcomes. The chart in Fig.5 shows the gas breakthrough of 12 wells which are not being monitored proactively have severely breakthrough of total 40% above their initial gas production. Root cause analysis revealed that these wells are all experienced a communication failure between the field devices and control unit.

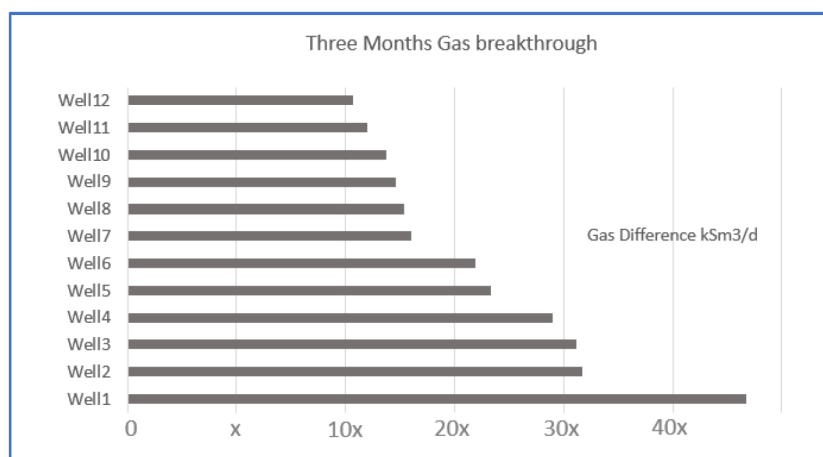


Figure 5—Hardware failures lead to value loss.

Lack of real-time gas breakthrough detection. Currently, the actual gas breakthrough indication is reliant on well testing, which operates on a lagging schedule. This delay can result in significant value loss, as gas breakthrough can persist for weeks or even months before a test is conducted and results are analyzed. Despite the availability of real-time data, such as the THP or GBC Differential Pressure (DP %) measurements, which could serve as early indicators of breakthrough, there is currently no dedicated platform or tool to leverage this data for real-time detection and alert generation.

Unstructured monitoring and performance tracking. The GBC optimization has been always on one direction, huddling around the optimum settings and then establishing one time result per optimization. The overall loop was not getting closed properly such as the compliance to these optimum setting over a period, compliance to hardware maintenance, even flagging the bad GBC actors was not established. This has led to loss the value of GBC technology and left many well in non-optimum operations.

People movement with lack of proper onboarding to GBC. People working with GBC technology must have a certain level of awareness about this technology. The technology itself is a combination of automation and reservoir knowledge. People movement has caused several breaks on the performance over the year. The situational awareness training was developed to onboard people to GBC, but it has not been consistent and not properly streamed.

Solution: Building a Structured Management System

In response to the previously outlined challenges, the 2024 GBC optimization campaign has shift beyond reactive fixes to a proactive, integrated system empowering data-driven decisions. The ambitious goal extends beyond achieving short-term gains, aiming for full digitalization of the process and a substantial leap forward in the technology itself. Given the complexity of the task, the approach has been broken down into manageable components, with a strategic emphasis on building the necessary infrastructure to support this ambitious goal. In the short term, the campaign continues to execute monthly targeted well optimizations while simultaneously improving processes and tools. This dual approach enables the team to implement and test improvements in real-time, ensuring that changes can be piloted and assessed immediately. Below are the steps taken and the resulted values.

Comprehensive Gap to Potential (GtP) Analysis

A detailed GtP analysis was conducted in early 2024 to assess the unrealized production potential across the studied fields. As shown in Fig. 6, the analysis indicated that optimizing underperforming GBC wells could free up approximately 1% of gas, potentially unlocking an additional 2% of oil production

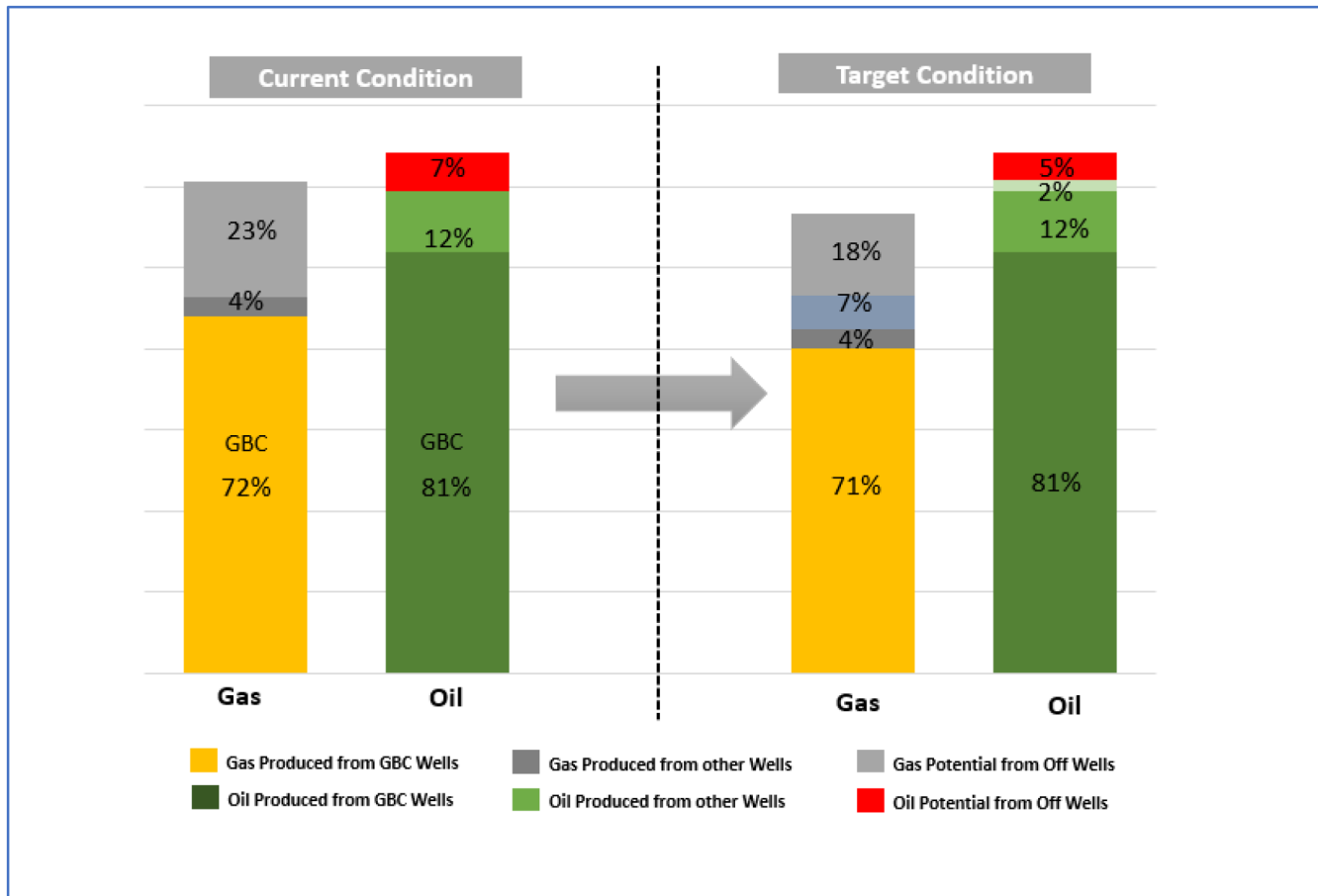


Figure 6—Gap to Potential analysis Jan 2024

The two bars on the left side present the total gas and oil at the current conditions (before the optimization exercise), while the two bars on the right indicates the gas and oil at the target condition. The GBC section indicates a 1% total gas reduction which is in actual figures is significant.

Developing a performance management system

Establishing a performance management system was a crucial step in transitioning from short-term, reactive actions to long-term, structured optimization. A comprehensive dashboard was developed to visualize well status, identify performance gaps, and facilitate stakeholder alignment. The dashboard is designed to serve three primary functions:

GBC Well Status Categorization. The large number of GBC wells necessitated a clear categorization of well statuses to prioritize actions and simplify stakeholder engagement. Figure 7 shows the categorization done for around 400 GBC wells, were classified into four categories:

- Optimized (14%) - Wells operating at their best potential.
- Shutdown (29%) - Wells requiring clarification from clusters and operations.
- Critical (28%) - Wells needing immediate optimization action.
- Maintenance (29%)- Wells under maintenance that may experience gas breakthrough if not addressed in time.

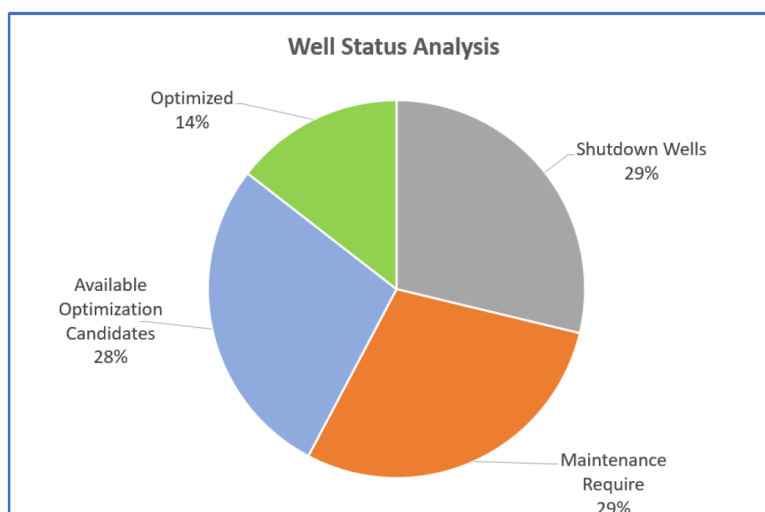


Figure 7—GBC Well Status Distribution

Real-Time Alerts for Top Bad Actors. The dashboard also includes a live alerting for wells exhibiting significant gas breakthrough. Wells are prioritized based on operational status. Fig. 8 shows a list of top bad actors, highlighting those in operation without maintenance restrictions, yet producing gas well above optimal targets.

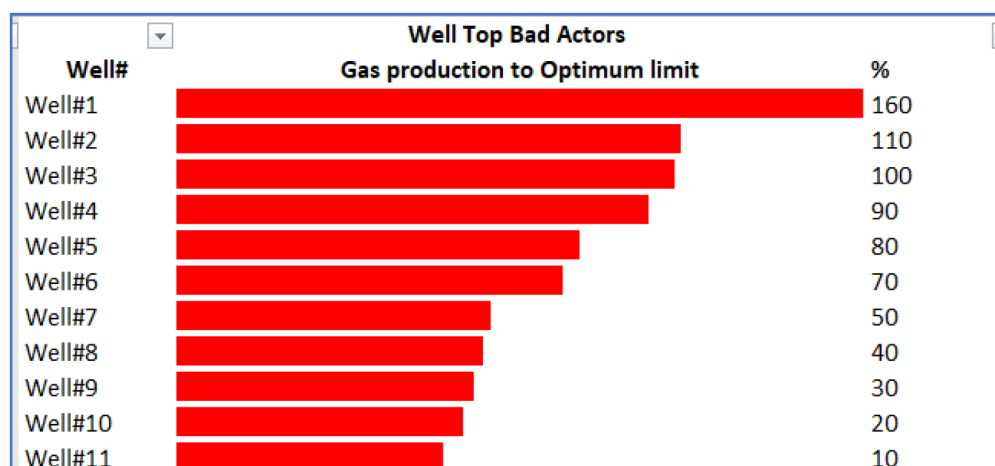


Figure 8—Top Gas Breakthrough Wells by Priority

Establish performance factors (KPIs) and tracking dashboards. As the optimization exercise is an ongoing process, a set of Key Performance Indicators (KPIs) was defined and visualized through interactive dashboards. By consistently tracking these performance factors, stakeholders can identify trends, detect deviations, and make informed decisions to ensure that the optimization efforts remain aligned with the intended objectives and deliver sustained value over time.

Fig. 9 illustrates the optimization outcomes and the value achieved for each individually optimized well, highlighting the reduction in gas production resulting from the optimization efforts.

Well# (dummy)	Optimization Trend	Target Limit (Offset)
WXX1		5.82
WXX2		2.65
WXX3		33.59
WXX4		-0.10
WXX5		2.83
WXX6		-7.32
WXX7		-9.61
WXX8		-3.53
WXX9		4.25
WXX10		-9.27
WXX20		24.13
WXX21		-9.32
WXX22		-13.84

Figure 9—Optimization outcomes in trends

Additionally, Fig. 10 presents the monthly progress of well optimizations, including cumulative implementation over time. It also shows the monthly well test performance, which is essential for assessing compliance with the optimization process—since, without well testing, the optimization loop cannot be fully closed.

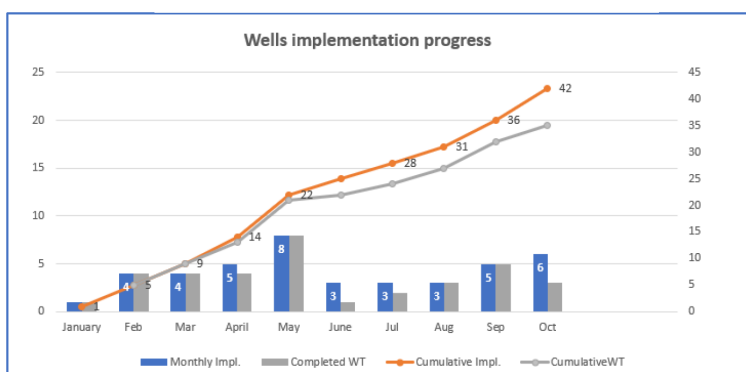


Figure 10—Monthly optimization progress

Finally, Fig. 11 presents the KPIs related to the optimization workflow, focusing on both performance and efficiency. Performance is measured by the implementation progress against the monthly targets, while efficiency reflects a combination of compliance with optimization targets and the completion of well tests.

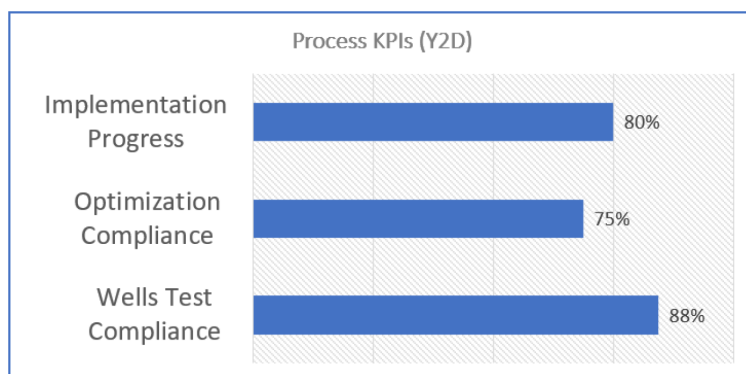


Figure 11—Work process KPIs

Establishing a Robust Optimization Work Process

The GBC optimization process has not been clearly defined in the past, as it was driven by a campaign-based approach that evolved frequently. Achieving a digitalized process was not feasible without first establishing a well-defined workflow. The workflow was progressively developed throughout 2024, with iterative improvements being tested and refined in real-time during ongoing campaigns. Fig. 12 presents the finalized and refined workflow, which has demonstrated effective outcomes in practice—including improved stakeholder engagement, streamlined action tracking, and consistent execution. The process begins with an initial performance review and the generation of optimization targets, coordinated centrally. These targets are then cascaded to the responsible parties—first to the Cluster Petroleum Engineers (PTs), followed by Operations (Ops)—to implement the recommended changes.

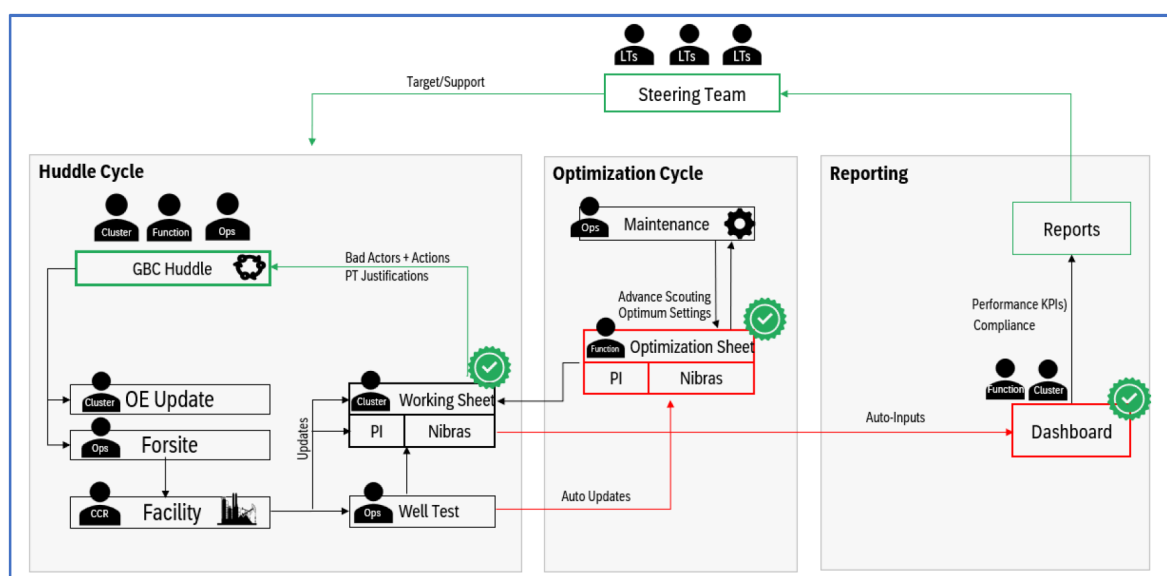


Figure 12—Structured Work Process

Subsequently, each well undergoes scheduled well testing to evaluate the results and close the optimization loop. An automated report is generated after each cycle to document outcomes and ensure traceability. At a strategic level, a steering committee reviews monthly progress, enabling high-level support, resource alignment, and issue resolution when needed.

Enhancing the optimization tools:

Traditionally, the optimization of Gas Breakthrough Control (GBC) wells was conducted on a well-by-well basis. Each individual well required a separate analysis using the optimization tool, with the process relying heavily on manual data input, multiple iterations, and team discussions to determine the appropriate GBC setpoints (SP) and gas lift (GL) recommendations. This manual workflow, while effective on a small scale, quickly became inefficient and unsustainable as the number of GBC wells expanded—especially in light of the rapid well count growth highlighted in Section 3.1.

Fig. 13 presents an example of the traditional optimization tool interface. The tool required multiple user interactions and stepwise input to converge on an optimal operating condition. This process had to be repeated for each well individually, consuming a significant amount of engineering time and creating inconsistencies in how optimization decisions were made across the field.

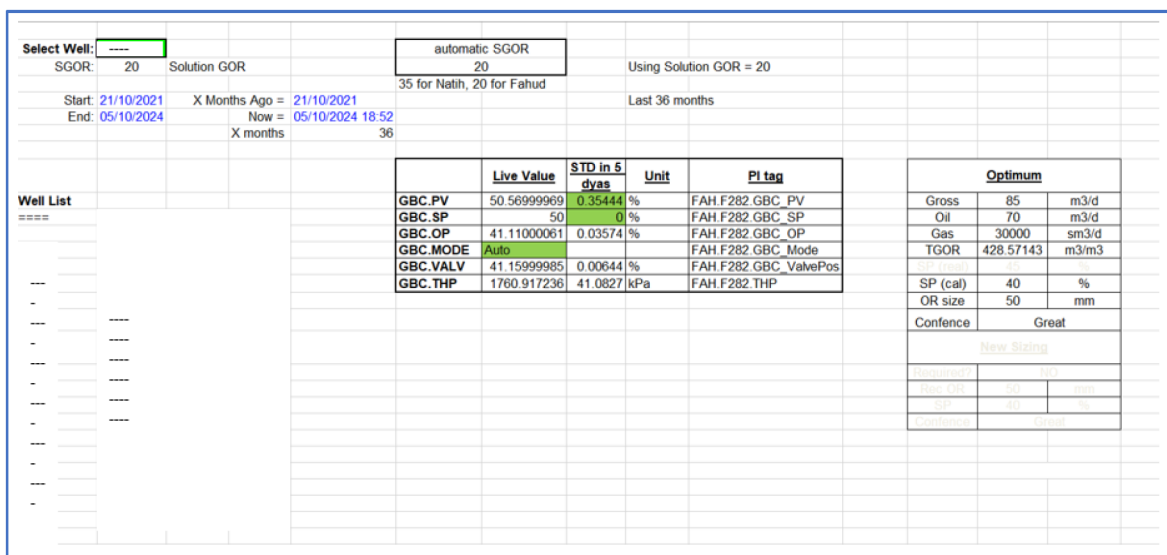


Figure 13—Single well optimization approach.

Recognizing the limitations of this approach, a new and enhanced optimization tool was developed as part of the digital transformation initiative. This advanced tool can retrieve and processing well test data for all GBC wells simultaneously. By centralizing and automating bulk calculations, the tool enables a holistic, field-wide view of well performance, supporting data-driven decision-making at scale.

The enhanced system also incorporates intelligent decision-support functions that help engineers identify wells requiring immediate attention, recommend optimal SP and GL targets, and prioritize actions based on performance metrics. Fig.14 illustrates this shift— from single optimization method to a comprehensive, bulk optimization approach. The figure highlights how all wells can now be visualized and analyzed together, with their operational status and optimization recommendations clearly presented in a single, user-friendly interface.

Wells Information		Latest Well Test Details								Live Analysis										
CONDUIT_NAME	STATION_CODE	E	GROSS	OIL	WATER	FORMAT_OIL_GAS	LIFT_GAS	Tot_GAS	TGOR	Suggested new GBC SP	Current GBC SP	SP at optimum?	Current GL	Target TGOR	Potential Oil	Potential Gas saving	Gas Lift Optimization ?	Last Well test Status	GBC Condition	Potential Gas Optimization
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---
---	---	2024	60.3303	8.19443	52.1359	25341.2	4996.62	30.338	3.698	40	32.39	NO	4,996.62	1,998.00	1.80556774	10337.78906	CUT GL	High Gas, Ok Oil	Man CNRT	YES
---	---	2024	58.4636	57.5792	0.88438	45952.3	384.554	46.337	805	45	50	NO	384.55	538.38	7.420784	11336.85156	seems OK	High Gas, Low Oil	Healthy	YES
---	---	2024	58.5139	58.5139	0	46270.3	2.18653	46.273	791	35	35	YES	2.19	583.24	1.4860611	11272.52734	seems OK	High Gas, Ok Oil	Healthy	YES
---	---	2024	38.4847	38.411	0.07375	40095.7	5707.17	45.803	1.192	25	60	NO	5,707.17	571.27	-3.410965	25802.85156	CUT GL	High Gas, Ok Oil	PV Issue	YES
---	---	2024	60.7624	20.1626	40.5998	24119.7	10059.2	34.179	1.694	25	40	NO	10,059.19	999.50	-0.1626434	14178.90625	CUT GL	High Gas, Ok Oil	Healthy	YES
---	---	2024	59.8769	39.6769	0	48106	0.32313	48.107	806	35	60	NO	0.52	437.45	20.3231392	13106.50781	seems OK	High Gas, Low Oil	Saturated	YES
---	---	2024	56.8239	56.8239	0	29388.8	15852.1	45.241	796	50	50	NO	15,852.11	454.46	-1.8239288	20240.89063	CUT GL	High Gas, Ok Oil	PV issue	YES
---	---	2024	19.1791	18.9396	0.23945	34960.8	-2922.52	32.038	1.691	70	80	NO	(2,922.52)	999.50	1.06037903	12038.28711	seems OK	High Gas, Ok Oil	PV issue	YES
---	---	2024	25.1938	24.9208	0.27301	46176.9	0	46.177	1.852	25	25	YES	-	999.71	10.079174	11176.90625	seems OK	High Gas, Low Oil	Healthy	YES
---	---	2024	66.5715	42.38	24.1915	50348	0	50.348	1.188	50	60	NO	-	999.71	-7.3799782	15347.96484	seems OK	High Gas, Ok Oil	Healthy	YES
---	---	2024	116.23	113.91	2.32	70398	0	70.398	618	40	60	NO	-	304.32	1.09	35398	seems OK	High Gas, Ok Oil	Healthy	YES
---	---	2024	116.995	44.4574	72.5378	47063.3	15212.5	62.276	1.400	80	90	NO	15,212.48	666.44	-14.457371	42275.82813	CUT GL	High Gas, Ok Oil	PV issue	YES
---	---	2024	76.8974	35.3093	41.5881	30004.9	14001.7	44.007	1.246	50	75	NO	14,001.71	1,199.52	-10.309299	14006.58203	CUT GL	High Gas, Ok Oil	Healthy	YES
---	---	2024	70.63	50.32	20.31	51342	0	51.342	1,020	40	70	NO	-	599.88	-0.32	21342	seems OK	High Gas, Ok Oil	Saturated	YES

Figure 14—Bulk Optimization tool

This development has significantly improved the efficiency of the GBC optimization process. Analysis and decision-making time have been reduced by approximately 80%, allowing the engineering team to respond more rapidly to performance issues and implement corrective actions across a large number of wells in a fraction of the time previously required.

In addition to time savings, this tool has enhanced standardization across the optimization workflow, strengthened the governance of GBC optimization, supported sustainable performance and aligning with PDO's broader goals for operational excellence and digital enablement.

Value Realization

By the end of 2024, the value generated from the newly adopted optimization approach proved to be significant and was widely recognized by key stakeholders. A total of 50 wells were successfully optimized, each demonstrating measurable improvement. These efforts resulted in a cumulative gas production reduction of approximately 1% of total gas production as depicted in Fig.15. This freed-up gas-handling capacity enabled the reactivation of lower-GOR wells, leading to an estimated increase of 2% in oil production. In addition, the campaign contributed to an estimated flare reduction of 3%, aligning with environmental and sustainability objectives. Operational efficiency also improved markedly, with an estimated 80% reduction in man-hour requirements due to streamlined analysis and decision-making processes.

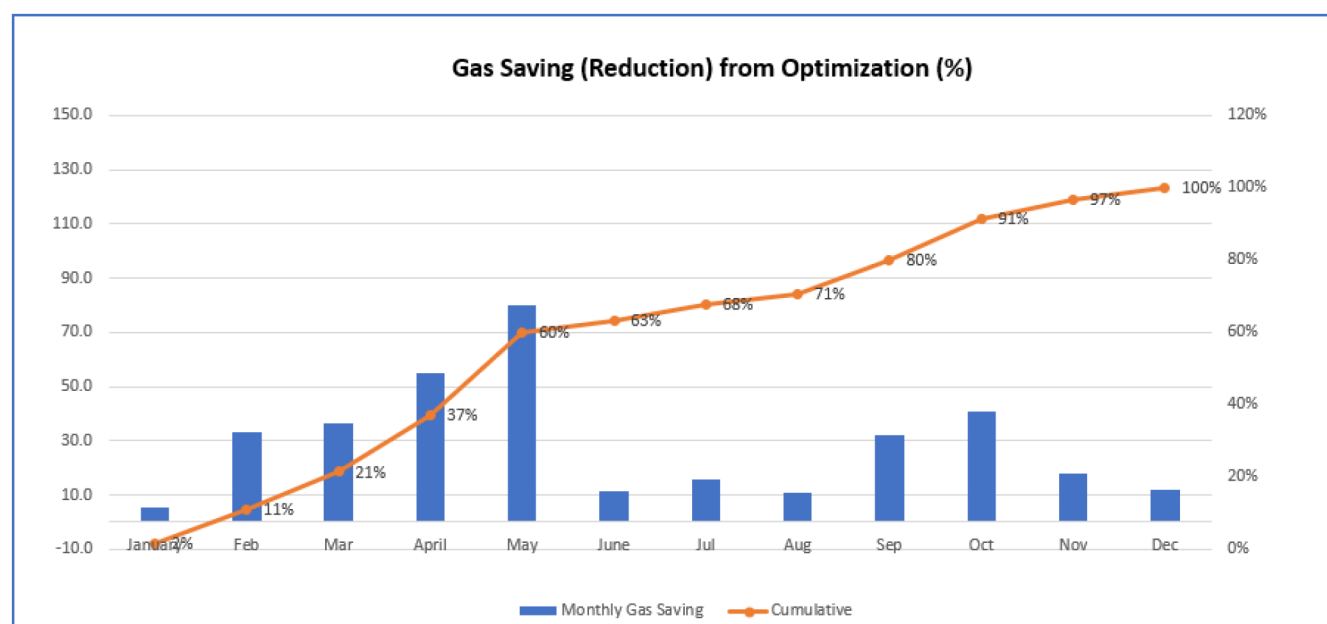


Figure 15—Achieved gas reduction through the year.

Future Strategy

The future strategy for GBC optimization centers on the full digitalization of the process, aiming to eliminate the traditional campaign-based model—including routine campaign huddles and leadership steering reviews. The goal is to transition from periodic, manually coordinated interventions to an automated, continuous optimization cycle, seamlessly integrated into the daily workflows of relevant stakeholders. As shown in Figure 16, the core of the digitalization will be the advance AI algorithm which developed to perform the below:

- Compute the optimization targets.
- Send the required actions.
- Receive the updates and validates the well tests.
- Analyze the status and generate reports.

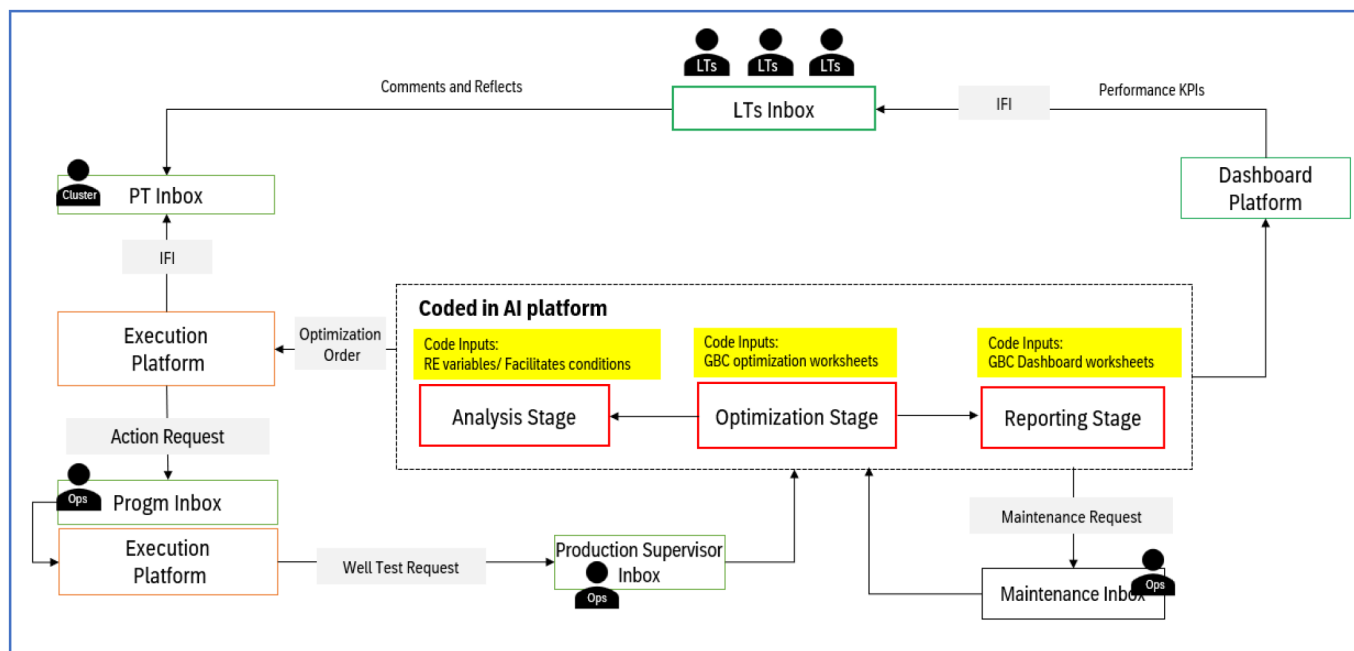


Figure 16—Digitalized work process

The workflow is designed to enable Petroleum Engineers (PTs), as the primary owners of production wells, to review and validate optimization settings prior to implementation. This step ensures a smooth transition from the current manual process to a fully digitalized system. Other stakeholders, such as programmers and operations teams, will act solely as executors of the recommended actions, with no input required from their side. However, rethinking the fundamentals as the long-term vision goes beyond digitalization toward full autonomy in GBC well optimization. With advances in real-time sensing, machine learning, and intelligent analytics, the need for periodic well testing and manual judgment can be eliminated. The goal is a self-optimizing system that continuously monitors production, adjusts control settings automatically, and enables engineers to focus on strategic oversight rather than routine tasks. This marks a transformative shift in how GBC wells are managed, unlocking higher efficiency and operational resilience.

Summary & Conclusion

The author has presented the transformation journey of Gas Breakthrough Control (GBC) optimization within PDO's major oil fields, transitioning from a reactive, campaign-driven approach to a proactive, data-centric, and digitally enabled framework. This shift was necessary to address long-standing operational challenges and support the growing scale of GBC wells.

The author has outlined the key building blocks that enabled this transformation, including the development of a structured management system, the implementation of performance monitoring dashboards, the automation of bulk optimization tools, and the standardization of workflows. These foundational elements helped establish a sustainable and scalable approach that reduces reliance on manual interventions and improves decision-making efficiency.

Finally, the author presented tangible results from the 2024 implementation, where 50 wells were optimized, leading to a gas production reduction of 1%, an oil production increase of 2%, and a flare reduction of 3%. Additionally, the process improvements achieved an estimated 80% reduction in man-hour effort.

This work highlights the significant value that can be unlocked through structured optimization and digital enablement, positioning PDO to further scale and sustain GBC performance across its asset base in the years to come.

References

1. Bruce Lundberg 2011, SPE 142617-PP: Well and Reservoir Behavior with Varying Associated Gas/Oil Ratio